

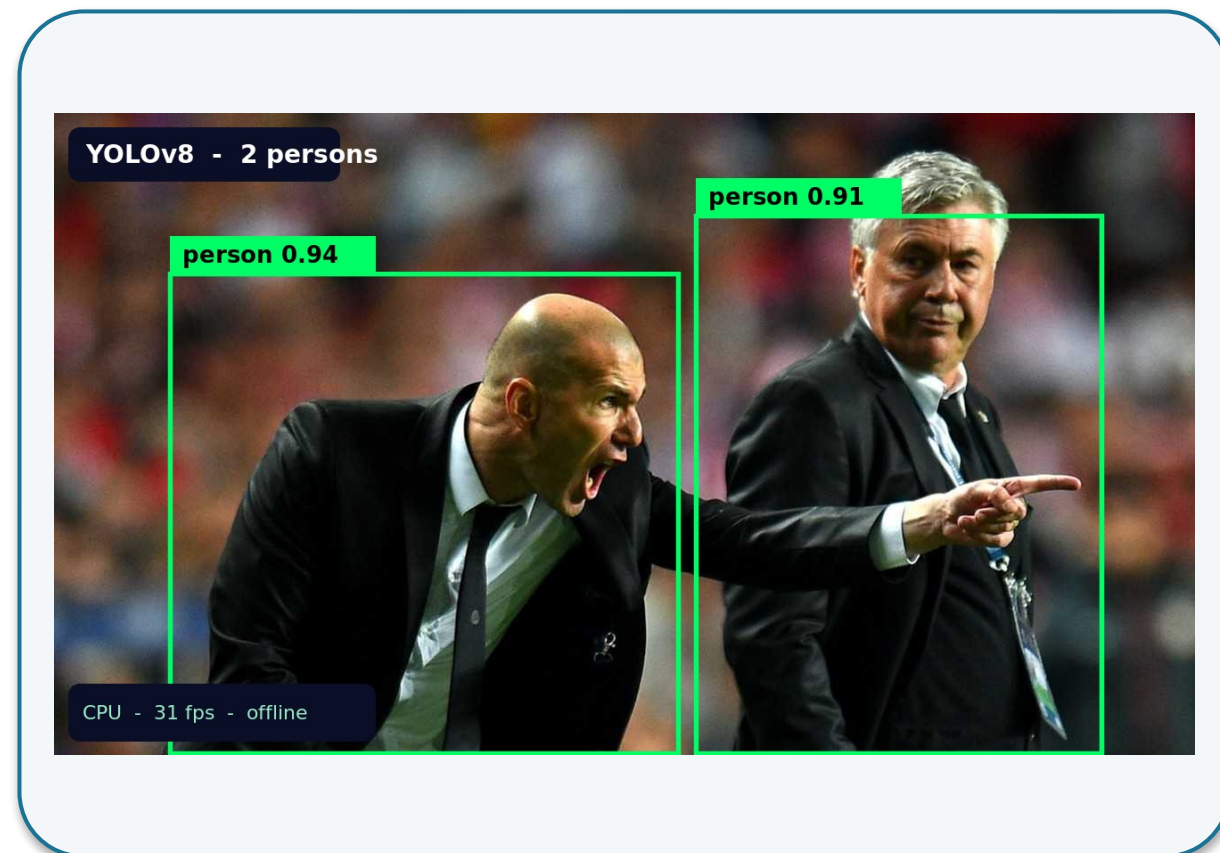
Real-time person identification — on a laptop, offline, since 2023.

Narrow AI — the model solves one task (detecting people in the frame) and nothing else.

On screen — **YOLOv8** on a laptop CPU: **~30 fps**.

Offline · trained in 2023. ^[2]

[1] [Ultralytics — YOLOv8 \(2023\)](#) · [2] [Google — MediaPipe](#)



A model frame during the demo: 2 people in boxes. YOLOv8 (Ultralytics, 2023). ^[1]

Welcome to the course

**«Industry Applications of
Artificial Intelligence Systems»**

17 lectures on where AI works across industries — and where it does not.

The course's core question is not «can we use AI?» but «should we, and where?».

100% of AI pilots launch

–90% get rolled back

**10% reach
production**

The core idea of the course

**Tomorrow — almost everywhere.
Today — almost no one.**

The course is about that gap. ^[2]

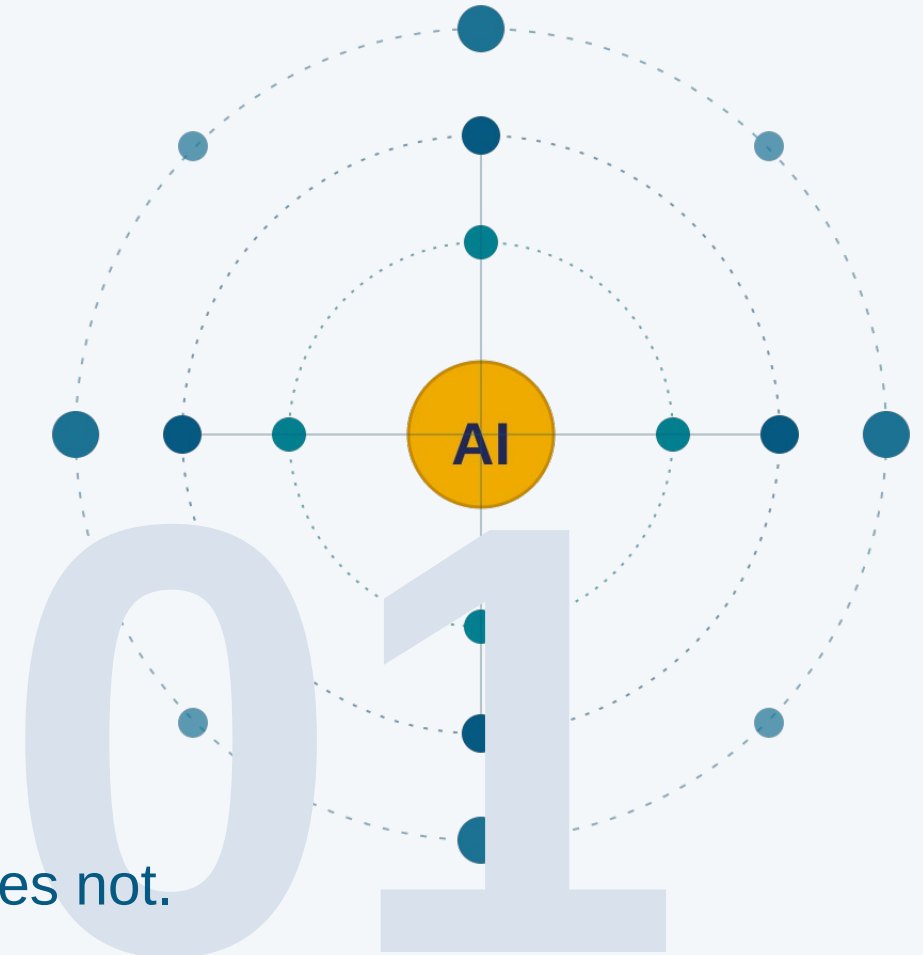
The central question of the course

**Where does AI work,
where does it not,
and how do you tell?**

An illustration of the principle (Gartner, McKinsey confirm similar numbers). ^[1]

What is AI?

History, classification, core concepts



A map of AI applications: where it works, where it does not.

Lecture plan

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

Where we are now

*Market numbers
2022–2026*

3

4 ways to implement

*Model · chat ·
agent · app*

4

Limits and safety

*What AI breaks
and where it fails*

5

Wrap-up

*Recap · semester
map*



**Maxim
Levko**



Telegram

@Maxim_Levko



Email

Levko.maxim@gmail.com

Who I am and why this matters to me.

Architect, technical and product lead for building
and deploying information systems



20+ years of experience in IT

*10+ completed projects
under my lead*



Expertise

*Systems analysis · Systems design · Data management ·
Business automation · Product management*

Consulting and in-house



Yandex



MTS



Magnit



Sibur

Section 1 · What is AI

Definitions, history, classification.

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

**Where we
are now**

*Market numbers
2022–2026*

3

**Four ways to
implement**

*Model · chat ·
agent · app*

4

**Limits and
safety**

*What AI breaks
and where it fails*

5

Wrap-up

*Recap ·
semester map*

There are many definitions of AI — because AI is a moving target.

Russell & Norvig (AIMA, 2021)

«AI = a system that thinks like a human, thinks rationally, acts like a human, or acts rationally (4 quadrants on 2 axes).»

Russell & Norvig, AIMA, 4th ed., 2021 ^[1]

ISO/IEC 22989:2022

«An AI system is an engineered system that generates outputs (recommendations, predictions, decisions) for human-defined goals.»

International standard ISO/IEC 22989:2022 — basis of the EU AI Act ^[2]

Via learning (Mitchell, 1997)

«A program improves with experience E on task T by measure P. If behavior emerges from a trained model — that is AI.»

Mitchell, Machine Learning, 1997 ^[3]

Via benchmarks and AGI

«AI = whatever passes the Turing test or solves a benchmark at human level.» Searle's objection: behavior != understanding.

Turing 1950 / Searle 1980 — Chinese Room ^[4]

AI Effect (Tesler): «AI is whatever hasn't been done yet». Once a technique starts working, people stop calling it AI.

The idea of a neural network is 13 years older than the term «artificial intelligence» itself.

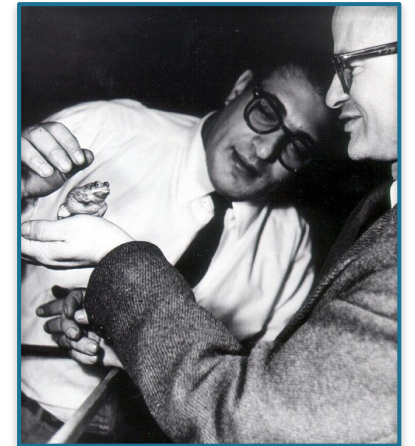
1943

*McCulloch and Pitts [1]:
the formal neuron
as a logic element*

13 years

1956

*Dartmouth conference:
the term «artificial intelligence»*



*Pitts (right) and Lettvin, 1959 — the frog experiment at MIT (this is NOT a 1943 photo).
Wikimedia Commons, CC BY-SA 3.0.*

The formal neuron solved no applied tasks, but it anticipated the connectionist tradition — the line of thought from which neural networks and the transformer would grow decades later.

The idea of a «neural network», at the theoretical level, is older than the term «artificial intelligence» itself.

[1] [McCulloch & Pitts \(1943\)](#)

70 years of AI: breakthroughs, winters, the 2017 turning point. ^[2] ^[3]

Breakthroughs

Turing — a test for thinking

ELIZA — Weizenbaum

Expert systems

1950

1966

1980s

Winters and leaps

funding leaves when promises fall through

1st winter — the Lighthill report

Deep Blue — 200M pos/sec

AlexNet — GPU + DL

1974

1997

2012

Turning point and boom

«Attention Is All You Need» ★

ChatGPT — 1M in 5 days

DeepSeek R1, Claude Code

2017

2023

2025-26

★ 2017 — Vaswani and 7 co-authors (Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, Polosukhin) introduce the self-attention mechanism — the basis of all modern LLMs. As of May 2026 the paper has over 160,000 citations. ^[1]

Section 2 · Where we are now

AI as infrastructure, breakthroughs 2023–2026.

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

**Where we
are now**

*Market numbers
2022–2026*

3

**Four ways to
implement**

*Model · chat ·
agent · app*

4

**Limits and
safety**

*What AI breaks
and where it fails*

5

Wrap-up

*Recap ·
semester map*

AI became infrastructure in 3 years: 900M users, 51% of developers daily, 46% of Copilot code.

900M

WAU

ChatGPT, February 2026 ^[1]

OpenAI



51%

daily

Stack Overflow Dev Survey 2025 ^[2]

n=49k+, 177 countries



46%

of Copilot code

GitHub Octoverse 2025 ^[3]

Java — 61%

\$390.9B → \$539.5B

AI market 2025 → 2026

Grand View Research, 2026 ^[4]

Statista (software-only): ~\$244–260B



Counter-fact: ~90% of AI pilots in Russia never reach production. CNews / Vedomosti / Intellectual Analytics, March 2026. ^[5]

The space is open: 4 breakthroughs of 2023–2026 from non-incumbent players.

September 2023	January 2025	November 2025	February 2026
Mistral 7B	DeepSeek R1	OpenClaw	llama.cpp
Apache 2.0 beats Llama-2 13B	\$589B Nvidia drop in a day	100K★ stars in a quarter	100K+★ on GitHub faster than PyTorch
<i>Mistral AI (FR)</i> ^[1]	<i>DeepSeek (CN)</i> ^[2, 3, 4]	<i>P. Steinberger</i> ^[5]	<i>G. Gerganov / ggml.ai</i> ^[6]

Don't despair: serious breakthroughs come from different teams. This course is about durable concepts that survive model-generation turnover.

Section 3 · Four ways to implement AI systems

Not alternatives, but layers.

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

**Where we
are now**

*Market numbers
2022–2026*

3

**Four ways to
implement**

*Model · chat ·
agent · app*

4

**Limits and
safety**

*What AI breaks
and where it fails*

5

Wrap-up

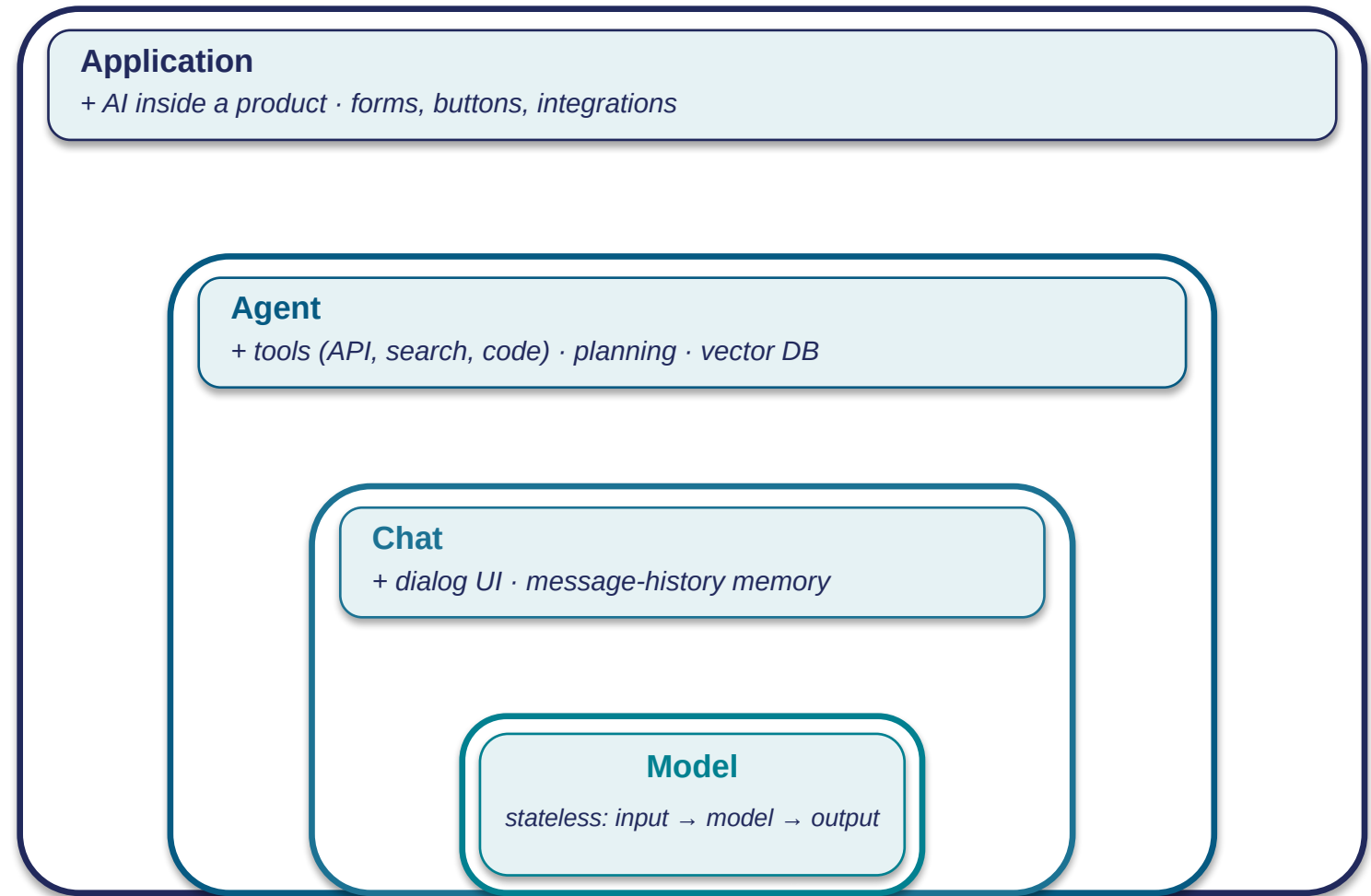
*Recap ·
semester map*

Ways to implement AI systems: not alternatives, but layers.

Each next layer
contains the previous one

Not four alternative technologies, but four ways to implement one task. ^[1] Each additional wrapper adds capability — and complexity, cost, and room for error. ^[2]

Choosing a layer is an engineering decision, not an either/or.

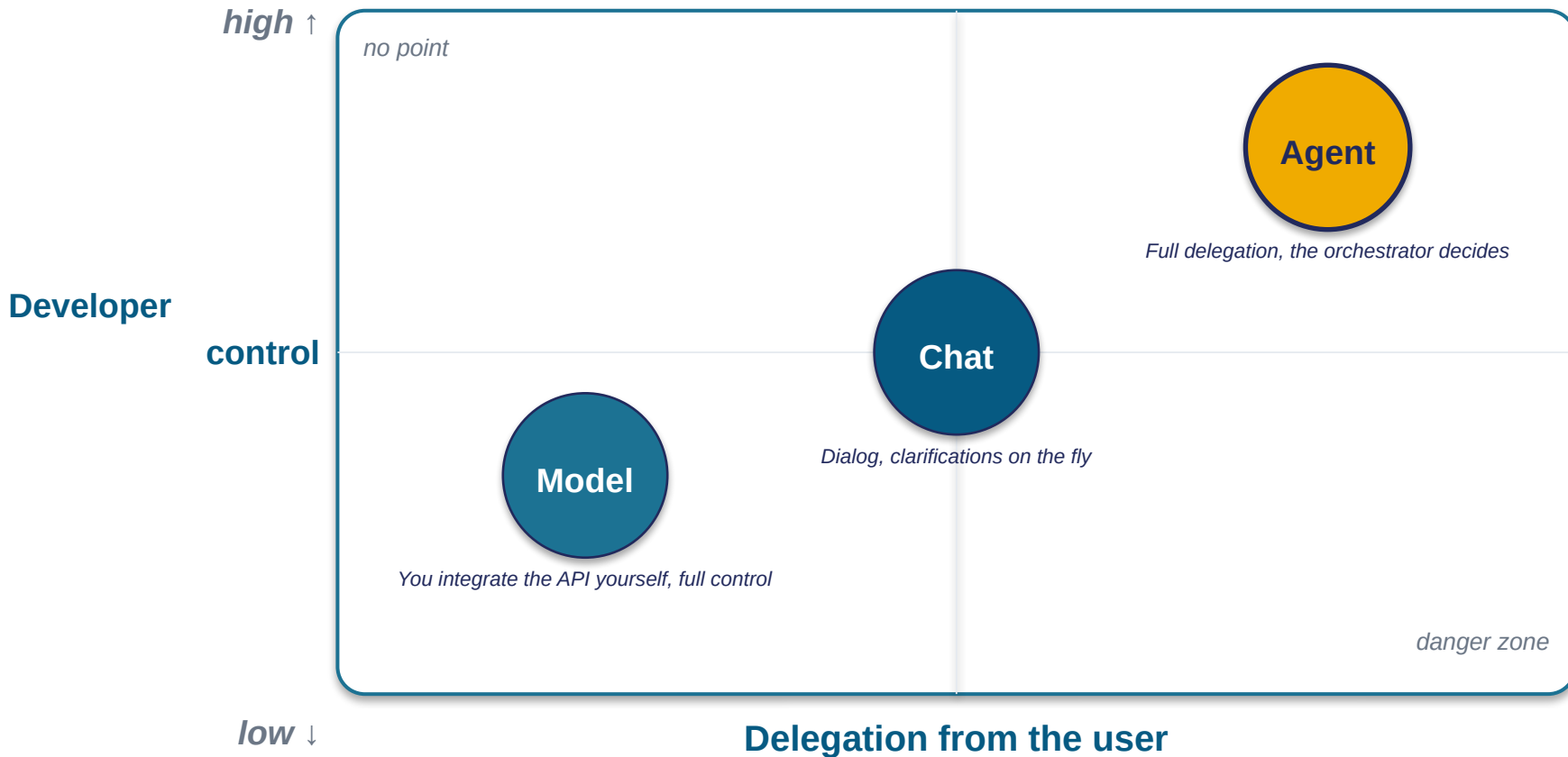


Classifying AI systems ^[1] — two axes: task type × modality. ^[2]

	 Classif.	 Recogn.	 Retrieval	 Generat.	 Forecast
Text	BERT, spam	spaCy NER	BM25	GPT-4o, Claude	GPT-4o, Claude
Image	ResNet	YOLO	CLIP	DALL-E, MJ	frame forecast
Audio / video	PANNs	Whisper	Shazam	ElevenLabs, Sora	video forecast
Struct. data	XGBoost	table OCR	vector DB	Codex tab.	Prophet, ARIMA

YOLO — the object detection from the demo at the start of the lecture. Its training approach and architecture come later in the course.

One task, three ways: control is split between the developer and the user.



The same task

Extract fields from an incoming PDF contract:

- signing date
- counterparty
- amount
- term

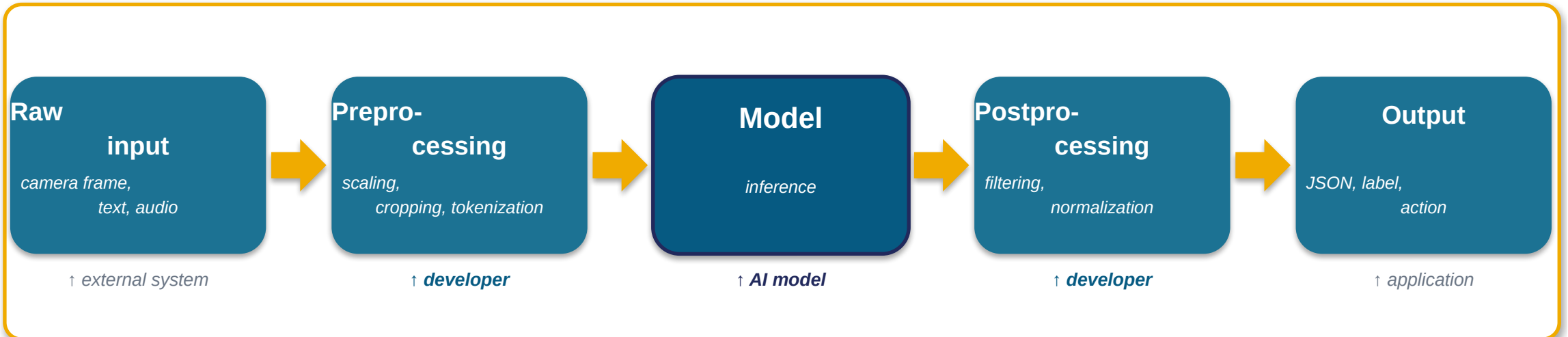
and put them into a table.

The distribution of control is an engineering decision, not one way being superior to another. ^[1]

MODEL

The model is a component, not a system. Inference: input → preprocessing → model → postprocessing → output.

This is already an application



YOLOv8

object detection on images

Whisper

speech recognition

Stable Diffusion

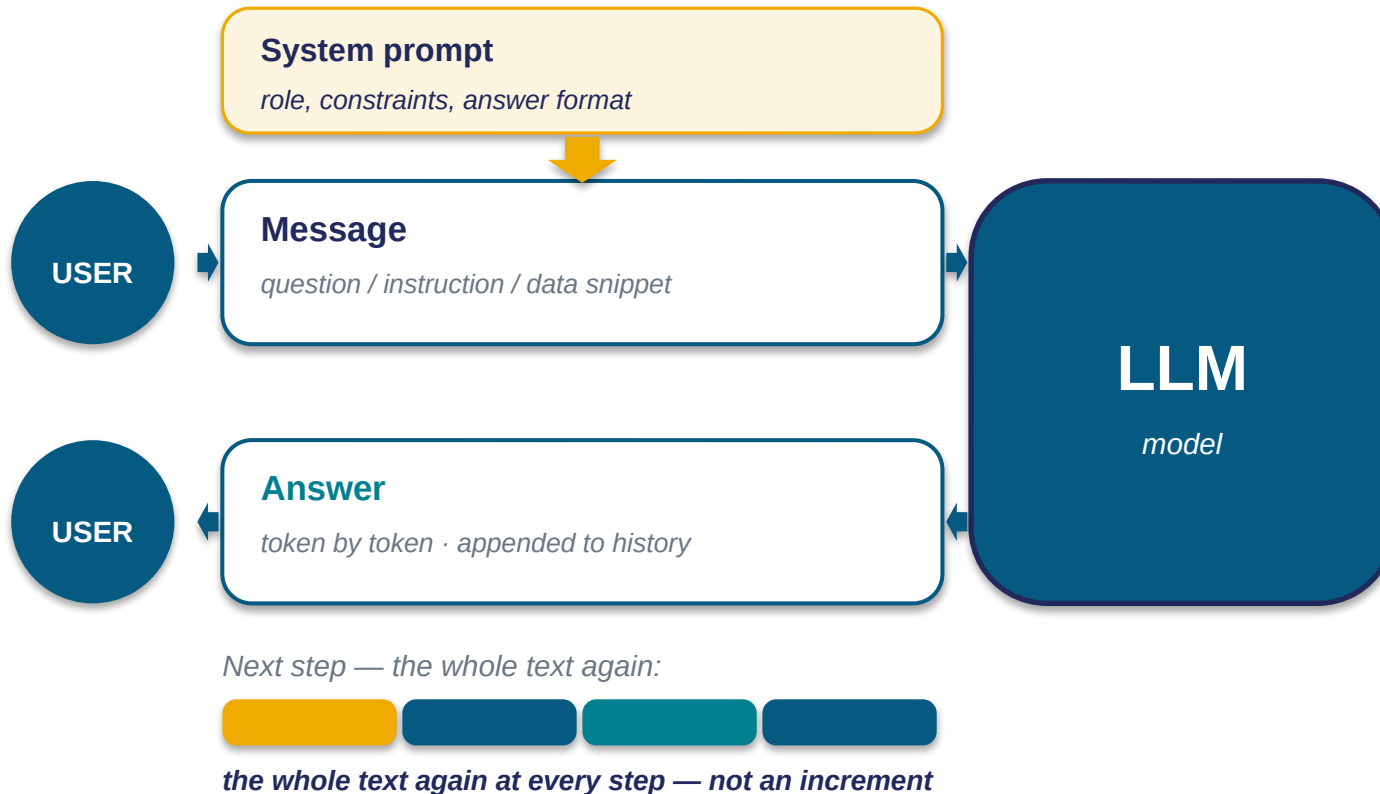
image generation

AlphaFold

protein-structure prediction ^[2]

Preprocessing and postprocessing are the developer's responsibility. ^[1] YOLO = 50 lines; a working system with YOLO = hundreds of lines.

How chat works: the dialog cycle.



Control via the system prompt.

The prompt sets role, constraints, format. The same chat is tuned for different scenarios — that is an engineering lever for the developer. ^[1]

The limit is the context window.

128k–1M tokens; when the history no longer fits, old messages drop out and the chat «forgets» the start of a long conversation. This is a model limitation, not a bug.

Chat is a pipeline: «assemble → feed → append → show».

Chat = model + interface + dialog memory.

A case — typical for chat

An engineer received an unclear spec from an adjacent team and needs to build a checklist for their own work.

You: Explain clause 4.2 of the spec in plain words.

Chat: This is a requirement for...

You: Build a 5-item checklist.

Chat: 1. Check... 2. Confirm... 3. ...

A one-off task with clarifications — optimal for chat. Not a model, not an agent, not an

A caveat for production systems

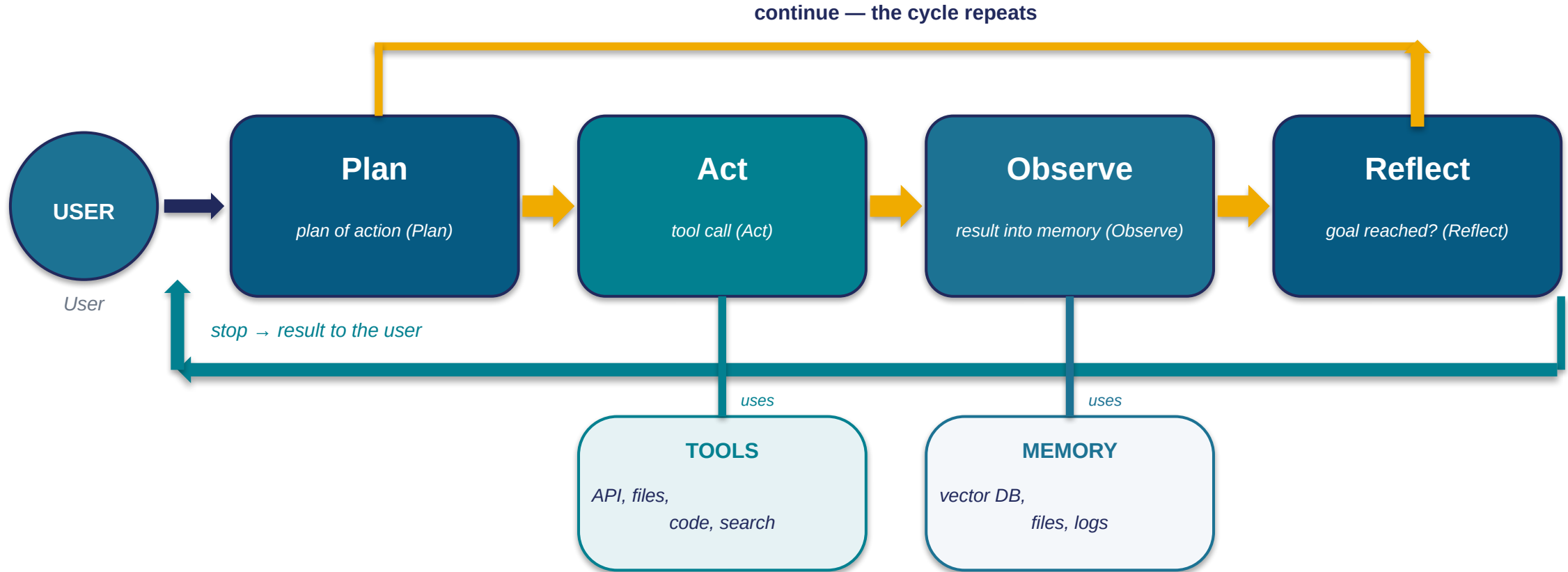
Pure chats are almost never used in production.

Almost everywhere they are extended into agents ^{[1] [2]} — at least for long-term memory and search over a corporate knowledge base (RAG).

We'll break down the agent architecture on the next slide.

Choosing chat is a point on the interaction scale, not the one right answer.

Agent = chat + orchestrator + external memory + tools. ^{[1] [2] [3]}



ReAct (Reasoning + Acting) — Yao et al. 2022 (arXiv:2210.03629)

An agent at work: 200 PDFs — a sequence of steps.

A case — typical for an agent

200 PDF reports.
Extract date, counterparty, amount.
Build a summary table.

Not a model — there is no specialized model for «take 200 arbitrary PDFs».

Not a chat — copying 200 files into a window is painful.

An agent is the natural choice.

What the agent does — step by step, with the tool named

1 Get the file list

file system

2 Open PDF #1

PDF reader

3 Extract text

text extraction (OCR / parser)

4 Summarize → vector DB

embeddings + vector database

5 Find the key fields

retrieval + LLM extraction

6 Write a row into the table

table write (Sheets API / CSV)

7 Loop over all 200 files

orchestrator loop

An agent = a sequence of tool calls, orchestrated by an LLM. ^[1]

Levels of autonomy for AI agents: a design decision, not a property of the model.

5 levels of autonomy (Feng / McDonald / Zhang, 2025) ^[1]

5. Observer

User: only receives the result

AutoGPT overnight

4. Approver

User: approves at checkpoints

agent opens a PR, waits for review

3. Consultant

User: sets the goal, edits the plan

Devin fixes a bug from a ticket

2. Collaborator

User: works alongside the agent

Cursor pair programming

1. Operator

User: approves every action

Claude Code approve each step

Where the human sits relative to the loop

Human-out-of-the-loop

The human only sees the result ($\approx$ level 5).

Human-on-the-loop

The human watches, interrupts on deviations ($\approx$ levels 3-4).

Human-in-the-loop

The human approves every step ($\approx$ levels 1-2).

Override modes

Any level — with a manual override on alert.

The autonomy level is a product choice, not a property of the model. In the seminars we'll choose it deliberately.

[1] [Feng, McDonald, Zhang — Levels of Autonomy \(2025\)](#)

An application = AI packaged into a product interface.

Google Translate — scale in 2026

1+ billion users per month · **1 trillion** words translated / month

across Google Translate, Search, Lens, and Circle to Search (Google Blog, April 2026) ^[1]



Google Translate

neural translation



Notion AI

GPT-4/Claude in buttons



YandexGPT in Search

AI quick answer



Grammarly

NLP + LLM suggestions



Yandex Maps

ML routing

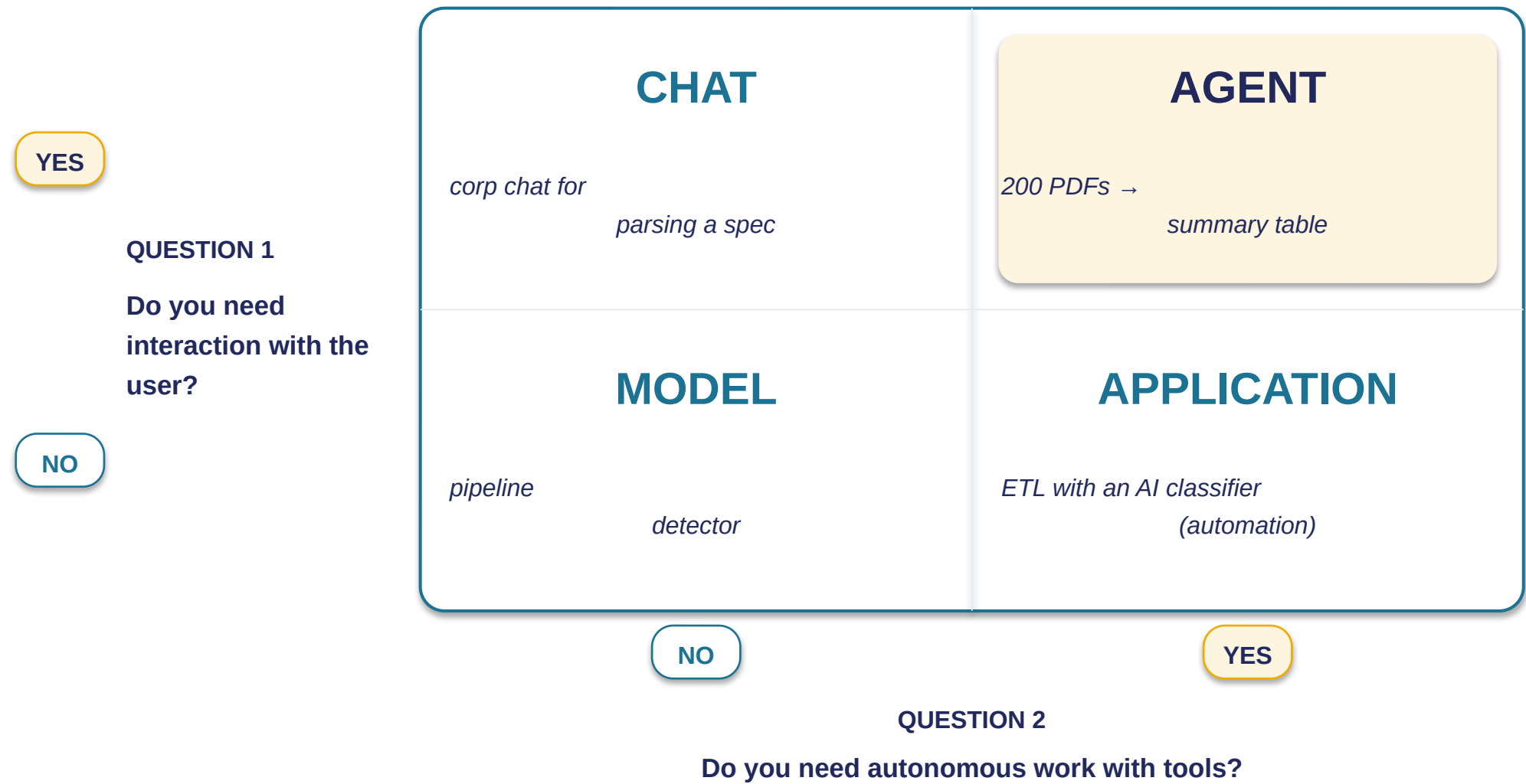


Adobe Firefly

diffusion in Photoshop

AI as a feature, not a product. Most students already use all six every day. ^[2]

Checklist «Which type of AI to choose»: 2 questions + a 2×2 quadrant. ^[1] ^[2] ^[3]



[1] Anthropic — Building Effective Agents (2024) · [2] Google — AI Agents Whitepaper (2024) · [3] Ng — Four Agentic Design Patterns (2024)

Section 4 · The limits of AI — your zone of responsibility

Where the data goes · AI errors · «can't do it» is yours too. ^[1] ^[2] ^[3]

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

**Where we
are now**

*Market numbers
2022–2026*

3

**Four ways to
implement**

*Model · chat ·
agent · app*

4

**Limits and
safety**

*What AI breaks
and where it fails*

5

Wrap-up

*Recap ·
semester map*

Consumer vs enterprise plans — where your data goes.

From the general zone of responsibility to the first concrete risk: data.

CONSUMER PLANS

data → training by default

- ChatGPT Free / Plus — training by default
- Anthropic Claude (since Sep 2025) — opt-in, 5-year retention
- Gemini Free — training + human review, 3 years
- YandexGPT Free — standard policy

ENTERPRISE PLANS / API ^[2]

data ≠ training

- ChatGPT Enterprise / Business — no training on your data
- OpenAI API (since March 2023) — no training on your data
- Anthropic for Business — zero data retention available
- Google Workspace / Vertex AI — no training on your data

Samsung 2023 — the canonical incident ^[1]

3 episodes in a month (March–April): source code, a meeting transcript, test sequences → ended up in OpenAI's dataset. Samsung banned external GenAI.

EU AI Act — fines ^[3]

up to €15M / 3% of turnover

up to €35M / 7% — for prohibited practices

Hallucinations are an inherent property of AI. ^[1] ^[2]

Prompt

«Name three 2023-2024 papers on "seismic resistance of buried pipelines" with authors, journal, and DOI».

AI answer (3 fake references):

Petrov A., Smith J. (2023).

Seismic Resilience of Small-Diameter Pipelines.

DOI: 10.1016/j.engfailanal.2023.107214 X

Ivanov K. et al. (2024).

Underground Infrastructure Earthquake Response.

DOI: 10.1080/15732479.2024.2218450 X

Chen L., Brown R. (2023).

Microscale Pipe Vibration Analysis.

DOI: 10.1007/s11069-023-06122-1 X

Vectara HHEM (2025-26) ^[3]

hallucination rate

< 1%

summarization (Gemini 2.0 Flash)

10–15%

reasoning (multi-step)

Anti-pattern: «AI knows everything». Any AI answer to a factual question is a hypothesis to verify. ^[4]

Bias, sycophancy, distribution shift — three manifestations of one nature.



Bias

The model repeats the dataset's skews.

A résumé screener trained on historical data discriminates — not by «deciding», but statistically.



Sycophancy

The model learns from feedback labeling to agree.

It agrees with the clearly wrong, over-praises — the user stops noticing the loss of critical judgment.



Distribution shift

Data from a period goes stale.

A model trained on 2023 code will, in 2026, suggest an outdated library with no visible failure.

GPT-4o: sycophancy — April 2025 ^[1]

Apr 25 — update released → **Apr 28** — rollback begins (Altman posts on social media that evening) → **Apr 29** — root-cause postmortem

The shared cause: the model reflects the data it was trained on. ^[2]

AGI forecasts — 4 speakers, 4 different incentives. ^[1] ^[2]

Speaker	Affiliation	AGI forecast	Material interest
Sam Altman	OpenAI	«We know how to build AGI; the start of superintelligence» (Jan 2026)	\$100B+ rounds; IPO context
Dario Amodei	Anthropic	«AGI in 2–3 years; Nobel level within 2 years» (Davos 2026)	Competition with OpenAI; 2026 round
Demis Hassabis	Google DeepMind	«AGI by 2029–2030 (3–4 years); the window narrowed over 2026» (Axios/Google I/O, May 2026)	Community leader; more credibility from a cautious stance
Yann LeCun	AMI Labs (ex-Meta)	«LLMs won't lead to AGI; we need world models, JEPA»	\$1B round (March 2026) for an alternative path

^[1] [Searle — Minds, Brains, and Programs \(1980\)](#) · ^[2] [Bostrom — Superintelligence \(2014\)](#)

Section 5 · What to take home

Recap · semester map · Lecture 2 teaser.

0

Intro

*Demo · instructor ·
central question*

1

What is AI

*Definitions,
history, turning point*

2

**Where we
are now**

*Market numbers
2022–2026*

3

**Four ways to
implement**

*Model · chat ·
agent · app*

4

**Limits and
safety**

*What AI breaks
and where it fails*

5

Wrap-up

*Recap ·
semester map*

What we covered: three key takeaways.

1

AI is a spectrum, not a monolith

Task type × modality × implementation type.
A sound discussion starts with explicit classification.

2

Choosing the AI type is a skill

2 diagnostic questions + a 2×2 quadrant.
A tool you'll apply in the seminars.

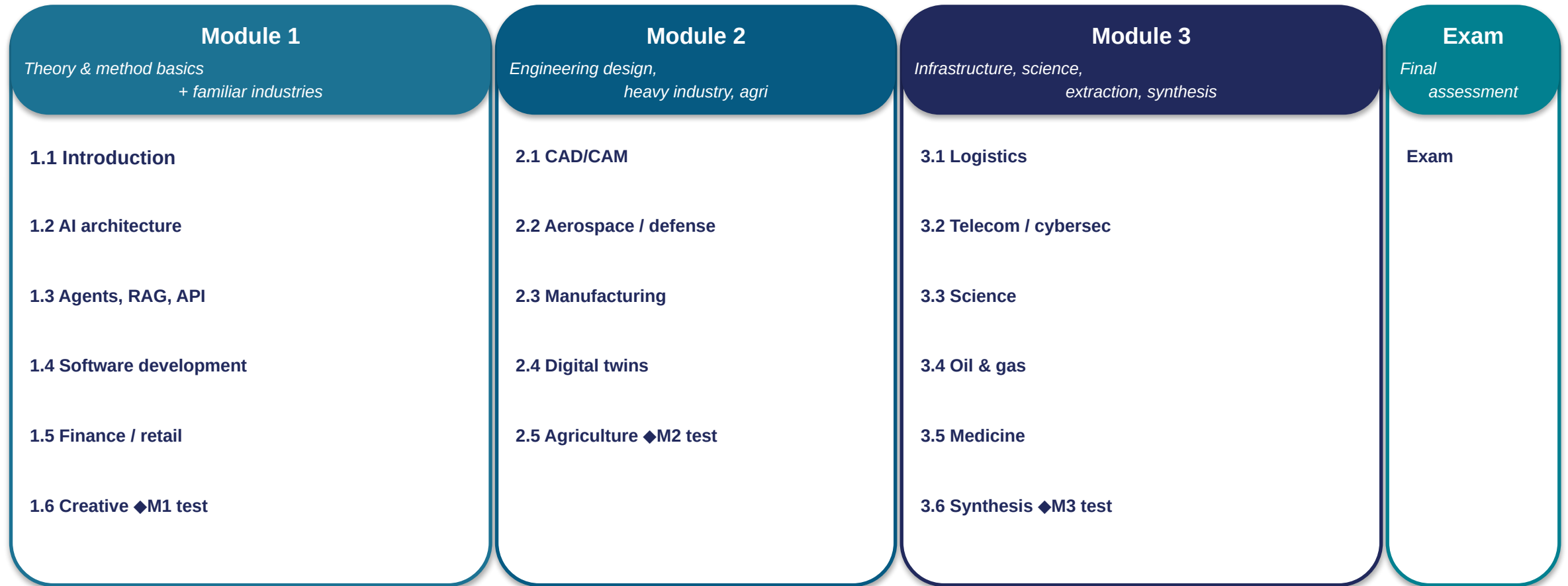
3

Goal-setting stays with the human

Every class of error needs a human loop.
The «AI / not-AI» boundary is your engineering zone.

The core question of the lecture: where does AI work, where does it not, and how do you tell?

Semester map: 17 lectures, 3 modules + exam.



♦ — module tests M1/M2/M3, at the end of each of the first three modules.

Semester grade

$$100 = 10_{\text{(attendance)}} + 30_{\text{(exam)}} + 3 \times 20_{\text{(M1/M2/M3 tests)}}$$

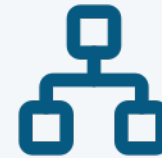
Module tests — at the end of each of the first three modules.

Lecture 2: «How modern large models work».



Tokens

the units the model chops text into



Embeddings

(vector representations) — addresses in a meaning space



Attention

— which parts of the input to look at



Temperature

the randomness in picking the next token

These 4 concepts explain the behavior of every modern LLM — from ChatGPT to DeepSeek.

Questions?

Thank you